A comment on “Cognitive causes of ‘like me’ race and gender biases in human language production” (Brough et al. 2024)

Marie Klünter[1 2], Rose Hörsting[3], Luca Leppert[1 2], Gina Reinhard[1 2] & Elen Le Foll*[1 2]

[1] Department of Romance Studies, University of Cologne
[2] CRC 1252 “Prominence in Language”, University of Cologne
[3] Institute of Linguistics, University of Cologne

* Corresponding author (elefoll@uni-koeln.de)

Abstract

The original study by Brough et al. (2024) investigates how social biases may influence automatic word retrieval mechanisms during sentence production. Across four experiments, the authors observe significant ‘like me’ effects, whereby speakers preferentially mention people who share their own race and gender before naming out-group members ($\beta = 0.16$, $SE = 0.02$, $z = 7.7$, $p < 0.001$). Response times are shown to increase as demographic match declines ($\beta = 0.02$ $SE = 0.005$, $t(642) = 4.5$, $p < 0.001$). Psycholinguistic manipulations show that the effect does not vary across grammatical constructions ($\chi^2(2) = 0.65$, $p = 0.721$), which the authors interpret as an indication that this bias arises from greater memory accessibility of in-group referents.

We conducted a computational reproduction using the authors’ original raw data and R scripts, supplemented by reconstruction of some missing scripts/steps, and performed multiple robustness checks (alternative model specifications, alternative exclusion criteria, etc.). Full reproduction of the original statistics was achieved for Studies 1–3 (identical point estimates, minor deviation in bootstrapped 95 % CIs). The statistical results of Study 4 were also reproduced except for a sign error. All figures were reproduced.

Full recreate reproducibility was not possible due to a lack of a README file, incomplete codebook, and undocumented analytic choices. We also noted several notable undisclosed deviations from the preregistered analyses. We struggled to map some of the reported analyses with the preregistrations linked in the article. In particular, although not labelled as exploratory, the Omnibus Study, which substantiates the main claim of the paper, appears not to have been preregistered.

We conclude that the findings reported in Brough et al. (2024) are computationally reproducible using the authors’ scripts with some minor adjustments and additions. We also found them to be robust to our modifications. To facilitate future repetitive research and cumulative endeavours, we recommend that multi-study papers provide detailed README files that clearly map individual files and preregistrations to each study, variable codebooks, numbered and annotated scripts, as well as explicit labelling of exploratory versus confirmatory analyses.

1. Introduction

In “Cognitive causes of ‘like me’ race and gender biases in human language production”, Brough et al. (2024) describe how speakers tend to first mention individuals who resemble them or their social group. The paper claims that this ‘like me’ effect is due to “(...) automatized mechanisms by which humans retrieve words from memory to produce sentences” (p. 1706). They demonstrate that the names of individuals who are more similar to the participants are more accessible to the participants.

The focus of the paper is the influence of gender and race. The effect was observed in four different studies. Testing both English and Chinese L1 and L2 speakers, the authors claim that the ‘like me’ effect is independent of demographic background. In Studies 1–3, participants had to learn the names of eight persons shown on photographs, who were either Black or white and male or female. They then had to describe events in which two of the persons, differing in either gender or race, engaged in symmetrical interactions. The aim of the studies was to measure “the degree to which participants framed their descriptions in ways that highlighted their own social group, by beginning their sentences with the name of the person who matched them in gender or in race” (p. 1707).

For Study 1, 60 Black women, 60 white women, 60 Black men, and 60 white men from English-speaking countries were recruited on Prolific. The participants first had to complete a memory task with 100% accuracy (learning to recognise eight named people from photos). This memory task was repeated every 20 trials throughout the experiment. In the main task, they saw the photos and produced a sentence to describe them following a given sentence frame. In Study 1, the ‘like me’ effect was observed (“Black women: 52.3% (50.1–54.9), $\beta = 0.12$ (0.06), $z = 2.1$, $P = 0.036$, Black men: 56.6% (53.8–59.8), $\beta = 0.28$ (0.07), $z = 3.9$, $P < 0.001$, white women: 53.6% (51.1–56.2), $\beta = 0.15$(0.06), $z = 2.6$, $P = 0.010$, white men: 57.3% (54.8–59.8), $\beta = 0.33$ (0.06), $z = 5.7$, $P < 0.001$), gender trials only (55.6% (53.6–57.9), $\beta = 0.29$ (0.05), $z = 5.4$, $P < 0.001$), race trials (54.2% (52.7–55.6), $\beta = 0.19$ (0.04), $z = 5.5$, $P < 0.001$”; p. 1707). The syntax of the sentence frame (transitive vs. intransitive sentences) had no significant influence in this study ($\beta = 0.03$ (0.06), $z = 0.4$, $P = 0.676$, p. 1708).

Study 2 was similar in procedure and replicated the findings of the Study 1 with new stimuli and participants, except that the ‘like me’ effect was not significant for Black women (51.9% (49.1–55.4), β = 0.09 (0.06), z = 1.5, P = 0.128, p. 1708).

In Study 3, 240 US-American participants were instructed to either form a passive or active sentence to describe the images. The procedure was similar to Study 1 and 2. The authors expected that in passive sentences the in-group member would be mentioned second. An overall ‘like me’ effect was observed, but it was not found to be significant for Black women or for race trials alone (Black women: 51.9% (49.3–54.5), β = 0.10 (0.06), z = 1.6, P = 0.106, race trails alone: 51.2% (49.6–52.9), β = 0.06 (0.04), z = 1.6, P = 0.096, p.1709). Another finding was that, contrary to the authors’ expectations, voice (active vs. passive sentences) had no significant influence on whether a ‘like me’ effect was found.

Study 4 tested whether the ‘like me’ effect could be observed when 120 participants used both their native language (Chinese) and an additional language (English). The ‘like me’ effect was observed regardless of the language used (Chinese as native language (56.2% (53.8–59.0), β = 0.23 (0.06), z = 4.1, P < 0.001), English as an additional language (53.7% (51.7–55.6), β = 0.13 (0.05), z = 3.0, P = 0.002), difference depending on the language used (β = −0.05 (0.04), z = 1.3, P = 0.192, p. 1710).

The data from Studies 1-4 were combined to find out whether the size of the like-me effect varied across groups. First, looking at race trials only, an effect was observed (β = 0.16(0.02), z = 7.7, P < 0.001). The effect was found to be slightly stronger among Asian women (β = 0.27 (0.11), z = 2.4, P = 0.0183, p. 1710). The ‘like me’ gender effect was significantly smaller in women than in men (β = −0.2 (0.05), z = 4.2, P < 0.001, p. 1710) and there was no significant ‘like me’ effect among Black women (M = 50.5 (48.3–52.7), β = 0.02 (0.05), z = 0.49, P = 0.623, p. 1710). The comparison of the ‘like me’ effect for race and gender across all four studies showed that the effect was overall smaller for race (M = 53.7% (52.8–54.6)) than for gender (M = 55.1% (54.1–56.1), β = 0.04 (0.02), z = 2.1, P = 0.0356 p. 1710).

One Rating Study measured intuitions about agency, while the other measured intuitions about speaker empathy and identification (p. 1713). For the former study, 64

new British English L1 participants were recruited. The names from Studies 1-4 and symmetrical verbs were used. The participants had to rate the two characters in the sentence in terms of volition and causation on a scale from 1-7. In the second Rating Study, 128 British participants read a mixture of active transitive, conjoined intransitive, and passive sentences. They subsequently rated the extent to which they thought that the speaker of these sentences identified more with the person mentioned first or second on a scale from 1–7.

Although the overall 'like me' effect was small, with mean values of around 55% (p. 1711), the authors conclude that the observed social bias is important. They suggest that the effect is independent of language, linguistic construction, and social group, although Bough et al. (2024: 1711) acknowledge that the first conclusion is based on just two languages (English and Chinese). According to the authors, "the fact that the like me effect was invariant across constructions suggests that it is driven by accessibility and incrementality, rather than by evaluation" (p.1711). They claim that "a core conclusion from the present studies is that the incremental and accessibility-driven language system that allows speakers to produce language that is fluent also leads them to produce language that is biased" (p. 1711).

2. Computational Reproducibility

The data and code were made available on the OSF (https://osf.io/87vew/overview). The repository includes raw and analysis data, as well as cleaning and analysis code in R. As the analysed paper comprises a multitude of studies, the total number of subdirectories and files was somewhat overwhelming. The absence of a ReadMe file made navigating the data packages challenging.

The preprocessing scripts for Study 1–3 (Process_Study1_Data.R, Process_Study2_Data.R, Process_Study3_Data.R) also mention that, due to a technical issue involving an unbacked-up computer, the authors no longer have access to all the initial raw data for these studies. However, they provide full access to data files, which have apparently only been hand-corrected for minor spelling mistakes (e.g. 'Rth' instead of 'Ruth'). We therefore classified these files as "raw data provided".

	Fully	Partial	No
Raw data provided	x		
Cleaning code provided	x		
Analysis data provided	x		
Analysis code provided	x		
Reproducible from raw data		x	
Reproducible from analysis data		x	

We were able to computationally reproduce the results of Studies 1-3 with the code provided by the authors. The only noticeable difference was a slight change in the numbers of the 95% bootstrapped confidence intervals with a maximal difference of ±0.2, which we believe is due to the way random seeds were set in the R scripts.

In Study 4, we were unable to fully reproduce two processed data files, study5_chinese.csv and data5_chinese.csv, from the raw data using the corresponding preprocessing script (Data Preprocessing/Chinese Study/Process_Chinese_Data.R). The script returned an error because the column `TrialType` does not exist. The raw data does include a column entitled `trial_type` which, however, only contains a single unique value: “survey_text”. By contrast, the `Trial_Type` variable contained in the processed data file uploaded by the authors, data5_chinese.csv, has two levels, differentiating between “RaceTrial” and “GenderTrial”. We were unable to reproduce this variable on the basis of the raw data, but all other variables were successfully reproduced. We therefore had to use the study5_chinese.csv and data5_chinese.csv files provided by the authors to reproduce the statistical analyses based on these data.

Moreover, we were unable to reproduce the median time taken to complete inputting the “critical sentence”, which according to the paper is 9.5 s in the English condition and 8.3 s in the Chinese condition (p.1713). It was not clear to us what exactly was meant by “critical sentence” and how these could be identified in the data sets. Furthermore, we obtained a minor difference in the final sample size for women in the English condition for Study 4. The paper refers to 28 women (p.1713), but after excluding those

with slow RTs, the script “Process_Chinese_Data.R” contains a table with `condition`, `ppt_gender`, `list` and `n` in which only 27 English women are listed.

We were able to reproduce the results of the ‘like me’ effect reported as part of Study 4. However, we were unable to locate the code necessary to reproduce some of the values reported in the paper and therefore had to write the code ourselves to check these numbers. This was the case for size of the overall ‘like me’ effect, reported as β = 0.18 (0.04), z = 5.0, P < 0.001, which we successfully reproduced, as well as the claim that “the size of the effect did not significantly differ depending on the language used” (p. 1710) where we reproduced again the same statistics as the authors apart from one value: we obtained a negative number for the *z*-value (*z* = -1.299) whereas, in the paper, a positive *z*-value (*z* = 1.3) is reported (p. 1710). This is likely to be a typo – it does not change the claims made in the paper. As for the earlier studies, we also observed minor differences in the values of the 95% bootstrapped confidence intervals.

For the Empathy Rating Study, the preprocessing script (Data Preprocessing /EmpathyStudy/process_data.R) filtered participants but did not create the processed dataset used in subsequent analyses. Consequently, it was originally unclear to us whether the data used in the analysis script (Data Analysis/empathy_study_data.csv) matches the output of the preprocessing steps. However, we were eventually able to successfully recreate the dataframe used by the authors in their analysis, based on an adapted version of their preprocessing script, the information provided in the preregistration (https://aspredicted.org/7698v.pdf), and the raw data provided by the authors (Data Preprocessing /EmpathyStudy/full_empathy_data.csv).

The Agency Rating Study is based on two separate preregistered studies and data collections (one for transitive and one for conjoint constructions), both carried out on Prolific. We were able to successfully reproduce the combined processed dataset from the raw Prolific data using the preprocessing script provided (Data Preprossessing/Rating Study/process_rating_data.R).

We were also able to successfully reproduce all of the plots presented in the paper (see OSF_Analyses_reproduction.qmd).

On p. 1709, the authors write:

> As Fig. 2a shows, the sizes of the like me effects varied only minimally across the grammatical constructions that we had tested ($\chi2(2) = 0.65$, $P = 0.721$). Rated agency and rated identification, by contrast, strongly varied based upon the grammatical constructions (both $\chi2(2) > 200$, $P < 0.001$) and clearly dissociated from one another.

We were unable to reproduce the first chi-square value. Our understanding of the code is that the “Test of grammar” section (from line 1022 of the OSF_Analyses.R script) best corresponds to this analysis. This test also returns a non-significant *p*-value (0.4886), but a chi-square value of 2.4272. We were able to reproduce the second chi-square value in the quoted paragraph above as 201.99 for the rated agency (transitive and conjoint sentences) and 820.96 for the rated identification (referred to in the code/data as the Empathy Rating Study). Both correspond to *p*-values that are indeed < 0.001.

We were able to recreate the processed dataset necessary for the analysis of the memory naming task (studies1_2_3_memory.csv) based on the raw data using the script overal_memory_data_processing.R; however, we noted that this script – which is over 1000 lines long – appears to repeat many of the data processing steps completed in earlier data processing scripts. This is particularly true of the removal of trials and participants according to the exclusion criteria; the latter of which is achieved by filtering out hard-coded participant IDs, which were presumably identified via the visual analysis of the tables produced in the script. It also includes some interactive elements (where the user needs to change a variable multiple times to check various subsets of the data) and many repeated code chunks within the script, too, making it difficult to check for consistency. In spite of the presence of code comments, it is difficult to follow the logic of the various data wrangling steps. As delivered, this analysis script is not reproducible. The main result of this analysis could be reproduced ($\beta = 0.02$ (0.005), $t(642) = 4.5$, $p < 0.001$). The mean reaction times reported on pp. 1709–1710 could all be reproduced.

The bootstrapped confidence intervals differed slightly, but this is not surprising as no random seed was set in the overall_memory_analysis.R script.

For the Multi-Study analysis, which combines the data across Studies 1 through 3, the Methods section (p. 1712ff.), describes what is being investigated in that part of the paper:

> We analysed the effect of syntax across the first three studies using a regression of the form LikeMe ~ 1 + Grammatical Frame + (1|subject) + (1|event).
> [...]
> The main omnibus analyses used mixed-effects regressions of the form LikeMe ~ 1 + Participant Gender × Participant Race + (1|subject) + (1 + Participant Gender × Participant Race|event). Predictors were contrast coded, and separate models were fit to the race and gender trials. The analysis of trial type (using data from all four production studies) had the form LikeMe ~ 1 + Trial Type [Gender versus Race-focused] + (1 + Trial Type|subject) + (1 + Trial Type|event)" (p. 1713f.).

These regression models can be found in the provided code and match the descriptions in the paper. We successfully reproduced the results of these analyses. The same numbers can be reproduced by running the original code.

2.1 Discrepancies Between Pre-analysis Plan and Article

The first challenge in comparing the preregistered analyses with the analyses reported in the article consisted in matching the preregistrations with the relevant study and results. We begin this section by reporting on our understanding of these matches before proceeding with a study-by-study comparison of pre-analysis plans and reported analyses.

2.1.1 Overview of preregistrations

The article provides links to the following Experimental Study preregistrations:

- Study 1: https://osf.io/bfhrn (06 June 2019)
- Study 2: https://osf.io/t928m (10 February 2020)
- Study 3 (passives): https://osf.io/drb6f/overview (11 October 2021)

Study 4 (Chinese Study) was not preregistered and is reported as such in the paper (p. 1752). In addition, a secondary analysis of the memory naming task (that was part of Studies 1–3) was preregistered a few years later: https://aspredicted.org/jf9cn.pdf (17 November 2023)

The preregistrations for the Rating Studies are distributed across three documents:

- Agency Study, transitive condition: "Does mention order influence perceptions of agency in conjoined phrases?" registered on 10 December 2020 at https://osf.io/zp5d4/overview. The second hypothesis ("Based off the previous experiment, we also predict that participants will be more likely to rate the first-mentioned name as agentive if it is a man's name rather an a woman's name.") as well as the interaction between the gender of the first-mentioned name and participant gender were not tested in the available analysis scripts and not reported in the paper.
- Agency Study, conjoined condition: "Does mention order influence perceptions of agency?" registered on 23 November 2020 at https://osf.io/rh9sd/overview
- Agency Study, passive condition and Empathy Study: "Intuitions analysis for Like Me study" (Study 1: judgements of agency in passive constructions; Study 2: judgements of empathy in conjoined, transitive, and passive sentences), registered on 21 November 2023 at https://aspredicted.org/7698v.pdf

Although each preregistration document was clearly formulated, the distribution of the preregistered studies and conditions did not correspond to the organisation of the Rating Studies in the paper, making it challenging for us to reconstruct the mapping between the preregistered analyses and the reported results.

2.1.2 Pre-analysis plan of Study 1

For Study 1, we found that the analysis code matches the descriptions of the analysis plan in the preregistration (https://osf.io/bfhrn). Brough et al. (2024) created two subsets for each gender and race trials, computing four separate generalized linear mixed-effect

models (GLMM) with `syntax` as a fixed predictor. Although the exact syntax of these models is not provided in the preregistration, the code for the GLMMs reflects the description:

```
OverallLikeMe ~ 1 + syntax + (1+syntax|subject) + (1+syntax|event)
```

This model is applied to the overall data, to each gender and race trials and to all possible gender and race subsets (white man, white woman, black man, black woman). The preregistered inference criteria reads: “We will use the standard $p < .05$ criteria for determining if the regressions suggest that the results are significantly different from those expected if the null hypothesis were correct.” Given that a total of seven GLMMs were conducted on the same dataset or subsets of the data, we were surprised that no mention was made of any *p*-value corrections.

2.1.3 Pre-analysis plan of Study 2

For Study 2, the analysis code differs from the preregistration (https://osf.io/t928m) in several ways. The provided code replicates the same models from Study 1 for the new data collected with improved stimuli in Study 2. A generalized linear mixed model was built for the overall data, for each race and gender trials, and for the gender and race subsets. The Methods section for Study 2 in the paper (p. 1712) explains that the same regression models were used as in Study 1. The preregistration, however, describes a more complex modeling strategy. Several preregistered analyses, including valence effects, questionnaire-based regressions, interaction tests and secondary control analyses, were not implemented in the available code. For the primary analysis, the authors preregistered an iterative approach to model building, looking at predictors of using the dominant group (man or white person) first. They start by including participant race and gender (a), then trial type (b), and finally syntax (c) as predictors of first mention of the dominant group:

(a) Man First/White First ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event).

(b) Man First/White First ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] + (1 + [Gender Trial Type or Race Trial Type] | Subject) + (1 + Participant Gender + Participant Race | Event).

(c) Man First/White First ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] * Syntax + (1 + [Gender Trial Type or Race Trial Type] + Syntax | Subject) + (1 + Participant Gender + Participant Race | Event).

None of these models, however, are provided in the analysis code and they are not mentioned in the paper.

In terms of the 'Like me' analysis, the preregistered analysis plan specified mixed-effects logistic regression models including participant gender, participant race, trial type, syntax, and their interactions in a similar iterative approach:

(d) Like Me Gender/Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event).

(e) Like Me Gender/Race ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] + (1 + [Gender Trial Type or Race Trial Type] | Subject) + (1 + Participant Gender + Participant Race | Event).

(f) Like Me Gender/Race ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] * Syntax + (1 + [Gender Trial Type or Race Trial Type] + Syntax | Subject) + (1 + Participant Gender + Participant Race | Event).

However, the analyses in the provided code only include syntax as the sole fixed effect, with participant demographics and trial type addressed only via data subsetting rather than factorial modeling:

OverallLikeMe ~ 1 + Syntax + (1 + Syntax | Subject) + (1 + Syntax | Event)

For both of the described analyses, the authors preregistered additional models with valence condition as a fixed effect, along participant gender, participant race, trial type and syntax, as predictors of the man first, white first, Like Me Gender and Like Me Race biases. These models are neither provided in the analysis code, nor mentioned in the paper.

For the final analysis of Study 2, Brough et al. (2024) preregistered regression models to analyse questionnaire data that contains participants' attitude towards homosexual and interracial relations, using them as predictors of demonstrating Like Me Race bias. As far as we could see, these models are not in the provided code either. As with Study 1, neither the preregistration nor the code or the paper seem to make any provisions to correct for the multiple testing problem.

2.1.3 Pre-analysis plan of Study 3

For Study 3, the preregistration (https://osf.io/drb6f/) differs considerably from the conducted Study. As a first difference, the authors preregistered valence as an independent variable with the levels intimate, casual and negative. As far as we can tell, this variable is not mentioned in the article. The authors also preregistered "exploratory models for each of the valence conditions" which we were unable to locate in the code or paper.

The authors preregistered two regression models to test their primary hypotheses of "Man First" and "White First":

a) Man First ~ Participant Gender * Syntax + (1 + Syntax | Subject) + (1 + Participant Gender | Event)
b) White First ~ Participant Race * Syntax + (1 + Syntax | Subject) + (1 + Participant Race | Event)

Similar to Study 2, these models are not reported in the article and, as far as we could tell, nor are they implemented in the analysis code.

Regarding the 'like me' effect, the models reported in the article and implemented/ present in the code test whether participants show a 'like me' effect in gender/race trials. The authors preregistered mixed-effects regression models with gender and race as fixed effects to test whether the 'like me' effect depends on participant gender or race (that is, whether one gender shows a stronger 'like me' than the other gender/race) and, additionally, whether that effect is moderated by `syntax` (interaction of gender and syntax):

a) Like Me Gender ~ Participant Gender * Syntax + (1 + Syntax | Subject) + (1 + Participant Gender | Event)
b) Like Me Race ~ Participant Race * Syntax + (1 + Syntax | Subject) + (1 + Participant Race | Event)

However, these models were not reported in the article. Similar to Study 2, the analysis in the provided code includes `syntax` as the only fixed effect:

OverallLikeMe ~ 1 + syntax + (1+syntax|subject) + (1+syntax|event)

Instead, the authors performed and reported subgroup analyses for the groups of black women, black men, white women, and white men. As they preregistered, they expected that each participant would most likely mention the figure first that matches their group. The preregistration also includes preregistered analyses of questionnaire data which we were not able to locate in the analysis code.

2.1.4 Pre-analysis plan of the Rating Studies

Our recreation of the empathy analysis data on the basis of the preprocessing script (Data Preprocessing /EmpathyStudy/process_data.R) and the raw data (Data Preprocessing /EmpathyStudy/full_empathy_data) confirmed that the dataset was derived from the original 128 participants, with 19 participants who failed *2 or more* catch trials excluded. This differs from the preregistration (https://aspredicted.org/7698v.pdf), which specified excluding participants who answered *more than 2* catch trials incorrectly. No additional exclusion criteria appear to have been applied. This potentially differs from the preregistration, which stated that trials with response times more than 3 SD from the mean, participants whose self-reported gender did not match Prolific records, and participants who had reported not to have lived in the UK for the last five years would be excluded. While these additional exclusion criteria might not have been relevant or may have been applied earlier on in the process, the preprocessing code of the Empathy Rating Study does not check for them.

By contrast, all of these preregistered exclusion criteria are applied in the processing scripts for the Rating Studies featuring transitive and conjoint constructions, except that, as with the Empathy Study, the exclusion criteria for failed catch trials implemented in the preprocessing scripts is also > 2 rather than ≥ 2 failed catch trials as specified in the corresponding preregistrations (https://osf.io/rh9sd/ and https://osf.io/zp5d4/). The processing script for the passive Rating Study does not include checks for any potential exclusion based on demographics.

The main claims of the paper are based on comparisons of models made with aggregated data from the various Rating Studies. As far as we can tell, these were not preregistered.

2.1.5 Pre-analysis plan of the Omnibus Studies

For the Multi-Study analyses, the preregistration does not appear to match the actual methodology for the analyses conducted across the first four Studies. The preregistration plan on *AsPredicted* (https://aspredicted.org/jf9cn.pdf) describes a different experiment that is part of the paper, namely the study on participants' reaction times, where the authors measure whether the participants' response speed is predicted by whether the character matches the participant in either race or gender. The actual preregistration contains the following analyses:

> For each accurate 8AFC response with RT < 10s, we predict logged RT in a mixed effects model by
> 1) Whether the character matches the participant's race _and_ gender (coded 1 in a dummy variable) or not (coded 0), i.e.,
> lmer(log(rt) ~ OverallLikeMe + (1+OverallLikeMe|id)
> 2) Whether the character matches the participant's race i.e.,
> lmer(log(rt) ~ like_me_race + (1+like_me_race|id)
> 3) Whether the character matches the participant's gender i.e.,
> lmer(log(rt) ~ like_me_gender + (1+like_me_gender|id)
> 3) Whether the character matches the participant's gender and race exactly (coded 1), matches only one feature (coded 0), or matches no features (coded -1)

lmer(log(rt) ~ like_me_graded + (1+like_me_gender|id)

Although these analyses are conducted in a different section of the paper, the main omnibus analyses reported in the paper (see p. 1714 for model formula) appear not to have been preregistered.

3. Robustness Checks

3.1 Robustness check (Study 2): Running the preregistered models

The reported models in the paper for Study 2 differ considerably from the models that were preregistered. While the models in the code provided predict the overall 'like me' effect with syntax as a sole fixed effect for each of the gender trial and the race trial subgroup, the preregistered models aimed to model participants' gender and race as fixed effects, as well as trial type and syntax to the models (see 2.1). As a robustness check, we ran the preregistered 2*2 models on each the gender trial data subset and the race trial subset:

Like Me Gender/Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event).

In both cases, the maximal random-effects structure specified in the preregistration failed to converge. In line with the preregistration, which specified that random effects accounting for the least variance and interactions preventing convergence would be iteratively removed, we simplified the random-effects structure by first removing the random slopes for participant gender and participant race at the event level, since the event-level slopes have near-zero variances:

Like Me Gender/Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 | Event)

Despite these adjustments and the use of the "bobyqa" optimizer, the model continued to exhibit singularity warnings, indicating that the remaining random-effects structure was still unsupported by the data. Only after removing all random effects did the model converge. However, we deem this simplification as highly problematic, as it eliminates the ability to account for individual variance within participants and systematic

differences across verbs. Nonetheless, we used the converged model as an approximation to the preregistered analyses. To facilitate interpretation of the fixed effects and to allow comparison with the reported descriptive results, we computed estimated marginal means using the *emmeans* package.

The estimated marginal means for the model on the "GenderTrial" subset are reported in Table 1 and a pairwise comparison of the estimated marginal means are illustrated in Table 2.

Table 1 – Estimated marginal means for 'like me' effect in gender trials

Participant Gender	Participant Race	Estimated Probability	SE
Man	Black	60.2%	0.013
Woman	Black	48.9%	0.012
Man	White	53.6%	0.013
Woman	White	53.6%	0.012

Table 2 – Pairwise comparison of estimated marginal means for 'like me' effect in gender trials

Contrast	Odds Ratio	SE	z	p
Man, Black / Woman, Black	1.58	0.12	6.23	< 0.001
Man, Black / Man, White	1.31	0.10	3.60	0.002
Man, Black / Woman, White	1.31	0.09	3.72	0.001
Woman, Black / Man, White	0.83	0.06	-2.71	0.034
Woman, Black / Woman, White	0.83	0.06	-2.82	0.025
Man, White / Woman, White	1.00	0.07	-0.01	1.000

The estimated marginal means and pairwise comparison show clear differences between Black women and Black men. In particular, Black women were less likely to show a ‘like me’ effect, a pattern that is consistent with the results reported in the original paper. Brough et al. (2024: 1710) suggest that this may be a “consequence of Black women having specific and distinct experiences with androcentric language, or it may reflect differences in how social groups experience and demonstrate subjectivity”. At the same time, the present analysis reveals that Black men displayed the strongest ‘like me’ effect of all groups, with significantly higher probabilities than both white participants and Black women. This pronounced effect for Black men is, however, not discussed explicitly in the original article.

For the race trial subset, the preregistered mixed-effects model failed to converge as well. Convergence could, again, only be achieved by eliminating all random effects resulting in a fixed-effects-only logistic regression. The model output (Table 3) indicates that this model explains virtually no variance in the data (Tjur’s $R^2 = 0.000$). None of the fixed effects, i.e. participant gender, participant race, or their interaction, significantly predict the likelihood of showing a ‘like me’ effect on race trials. This suggests that, under the preregistered modeling framework, there is no detectable evidence for systematic variation in ‘like me’ biases on race trials across participant groups.

Table 3: Results of the preregistered ‘Like me’ model for race trials (convergence-adjusted)

	OverallLikeMe		
Predictors	*Odds Ratios*	*CI*	*p*
(Intercept)	1.21	1.15 – 1.27	**<0.001**
Participant Gender Woman	0.99	0.94 – 1.04	0.727
Participant Race White	1.00	0.95 – 1.05	0.956
Participant Gender Woman X Participant Race White	0.98	0.93 – 1.03	0.395
Observations	6602		
R^2 Tjur	0.000		

3.2 Robustness check (Study 3): Running the preregistered models for gender and race

As the preregistered models for Study 3 differ from the models reported in the article in that they include gender and race as fixed effects, we performed a robustness check by computing the preregistered models using the provided data.

For gender trials, the preregistered model yielded almost identical results to the model reported in the article. In both the preregistered and the reported model, participants chose the same-gender character first in 55.9% of trials. The intercept was significant, confirming an overall 'like me' effect in gender trials (i.e. both men and women were, on average, biased toward their own gender). Fixed effects (participant gender, syntax and their interaction) had no significant effects, in line with Brough et al. (2024: 1709)'s statement that there was no significant difference between men and women concerning their 'like me' effect and no difference between active and passive sentences.

The analogous preregistered race model, fit to race trials, showed no clear evidence for a 'like me' effect. Again, the values computed by the preregistered model match closely the ones reported by the authors (Table 4). Participant race and syntax were not significant, neither was their interaction, indicating no significant difference between White and Black participants and no difference between active and passive in race trials.

The maximal preregistered models only converged when using the "bobyqa" optimizer and produced boundary (singular) fit warnings, indicating that the random effects explain little to no variance. This may explain why the authors decided not to report the preregistered models. However, the preregistration stated that the authors would "iteratively remove random effects that account for the least variance and interactions preventing convergence". Following this protocol, we removed the random effect for event (both intercept and slope) as it explained least of the variance (0.01). Once this random intercept was removed, the models converged without the optimizer, suggesting that convergence issues are unlikely to be the main reason for not using the preregistered models for the analysis. Importantly, the simplified models allowed

computation of the conditional R^2 values which closely matched those of the reported models, while the fixed-effect estimates remained essentially unchanged. In both the reported and the preregistered models, marginal R^2 was very small (0.000 – 0.002), indicating that the fixed effects explain virtually no variance. In contrast, the conditional R^2 (0.17 – 0.20) shows that random intercepts and slopes by subject accounted for most of the variance accounted for by the model. This pattern is consistent with the presence of an overall 'like me' effect in gender trials, with little evidence that this effect is modulated by participant gender, participant race or syntax. In sum, implementing the preregistered model as a robustness check of Study 3 had no effect on the magnitude or the statistical significance of the main point estimate.

Table 4 – Study 3: 'Like me' effect in Preregistered Gender and Race Models

Model Trial Type	'like me' effect original study	'like me' effect preregistered models (robustness check)	'like me' effect preregistered models without random effect of event
Gender	M = 55.9% [53.9 – 57.7], β = 0.28 (0.05), z = 5.2, P < 0.001 Marginal R^2/ Conditional R^2: 0.000/0.199	M = 55.9% [53.8 – 57.6], β = 0.28 (0.05), z = 5.3, P < 0.001 Marginal R^2/ Conditional R^2: 0.001/ NA	M = 55.9% [53.8 – 57.6] , β = 0.28 (0.04), z = 6.3, P < 0.001 Marginal R^2/ Conditional R^2: 0.001/ 0.195
Race	M = 51.2% [49.6 – 52.9], β = 0.06 (0.04), z = 1.6, P = 0.096 Marginal R^2/ Conditional R^2: 0.000/0.199	M = 51.2% [49.6 – 52.8], β = 0.06 (0.04), z = 1.7, P = 0.086 Marginal R^2/ Conditional R^2: 0.002/ NA	M = 51.2% [49.6 – 52.8], β = 0.06 (0.04), z = 1.7, P = 0.085 Marginal R^2/ Conditional R^2: 0.002/ 0.167

3.3 Robustness check (Study 4): excluding neutral chinese names

In Study 4, the authors indicate that the observed 'like me' gender effects may be smaller in Chinese because of including two possibly gender neutral chinese names (霍普 [Hope] and 贝斯]Beth]). Therefore we decided to rerun the analysis for Study 4 with

a subset of the data that excluded these two neutral names. In Study 4, the authors only differentiated between the overall 'like me' effect and the overall 'like me' effect depending on "condition" (Chinese or English) and "demog" (men or women). Therefore, we first examined whether the overall 'like me' effect increases when neutral names are excluded. Indeed, when excluding neutral names, the 'like me' effect are larger than those reported in the paper (see Table 5). Next, we calculated the means for the 'like me' gender effect for the Chinese condition and observed a small increase, when neutral names are excluded, but it should also be noted that the 95% CI is wider than that with the neutral names.

Table 5: Study 4: 'like me' effect without neutral names

Dataset	**Including neutral names**	**Excluding neutral names (robustness check)**
Overall 'like-me' (condition = Chinese)	M = 56.2% of trials [53.8–59.0], β = 0.23 (0.06), z = 4.1, P < 0.001	M = 57.7% of trials [54.8-61.2], β = 0.290 (0.064), z = 4.544, P<0.001
Overall 'like-me' men	M = 54.7% [52.3–57.3]	M = 55.2% [52.4–58.1]
Overall 'like-me' women	M = 55.3% [53.2–57.4]	M = 56.2% [53.7-58.9]
Overall 'like-me'	M = 55% of trials [53.4–56.8], β = 0.18 (0.04), z = 5.0, P < 0.001	M = 55.7% of trial [53.8-57.8], β = 0.209 (0.037), z = 5.580, P < 0.001
'like-me' gender means, condition=Chinese (not reported in the paper)	M = 54% [52.1-56]	M = 55% [49.5-60.1]

3.4 Robustness check (Study 5): Memory naming task

We noticed that the main claim made on the basis of these data relies on a GLMM in which the 'like me' effect is modelled as a numeric predictor ranging from -1 to 0. However, this variable only has three possible values that, somewhat counterintuitively, correspond to the following:

- -1 = highest me-likeness (i.e. both race and gender match participant demographic)
- 0 (only race or gender matches participant demographic)
- 1 = lowest me-likeness (neither race nor gender match participant demographic

We believe that this variable would be better modelled as an ordinal categorical variable, hence we reran the reported model with the independent variable modelled as an ordered factor.

The coefficient of interest turns out to be robust to this manipulation. It is of a similar size than the reported coefficient and also statistically significant (see Table 6). Note that in both the model reported in the paper and this additional model, the LikeMeFeature predictor accounts for just 0.01% of the variance in participants' recall times. This statistic is not reported in the paper.

Table 6: Summary of model with ordered factor predictor variable

		log(rt)	
Predictors	*Estimates*	*CI*	*p*
(Intercept)	7.92	7.90 – 7.94	**<0.001**
LikeMeFeatureNumeric L	0.03	0.02 – 0.04	**<0.001**
LikeMeFeatureNumeric Q	-0.01	-0.02 – -0.00	**0.038**
Random Effects			
σ^2	0.18		
$\tau_{00\ id}$	0.07		
$\tau_{11\ id.LikeMeFeatureNumeric}$	0.00		

$\rho_{01\ id}$	0.11
ICC	0.28
N_{id}	622
Observations	22482
Marginal R^2 / Conditional R^2	0.001 / 0.283

In the original analysis of the memory naming task, the authors removed trials for which participants needed longer than 10 s to name the images (as preregistered). Below are the descriptive statistics of the response time variable for the trials that were included in this analysis (i.e. first guesses only, correct responses only, and excluding participants with strong biases). It turns out that 10,000 ms is much longer than most people needed to recall the names:

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
    298    1891    2627    3663    4040 1234104 
```

At the same time, we noted that the overall_memory_analysis.R script produces a plot which illustrates the difference in response time across the four conditions. This plot, which is not included in the final version of the paper, shows that the differences in mean response times are very small, that the confidence intervals overlap, and that there is considerable variation across the entire dataset (see Fig. 1). In fact, not all of this variation is visible as the plot cuts the *x*-axis off at 4,500 ms, thereby removing 366 data points.

When we visualise all the data points entered into the reported model as boxplots, a rather different picture emerges (see Fig. 2). The pattern in median response time does not match the reported effect.

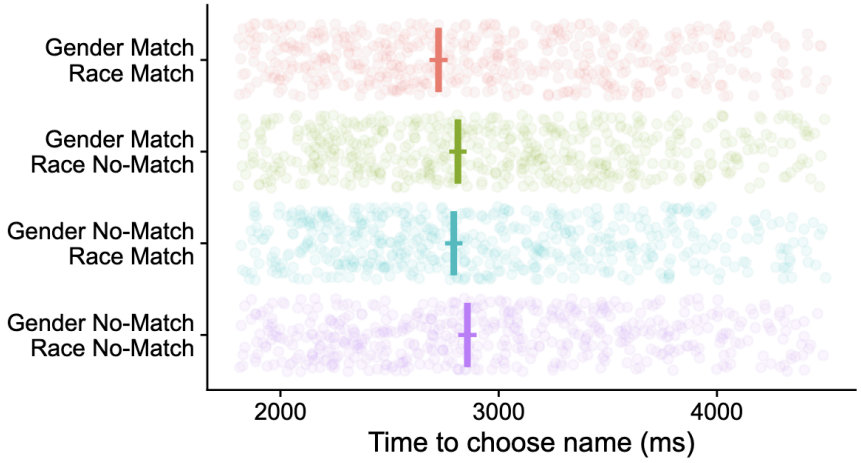


Fig. 1: Visualisation showing response time across the four conditions (as coded in original analysis script)

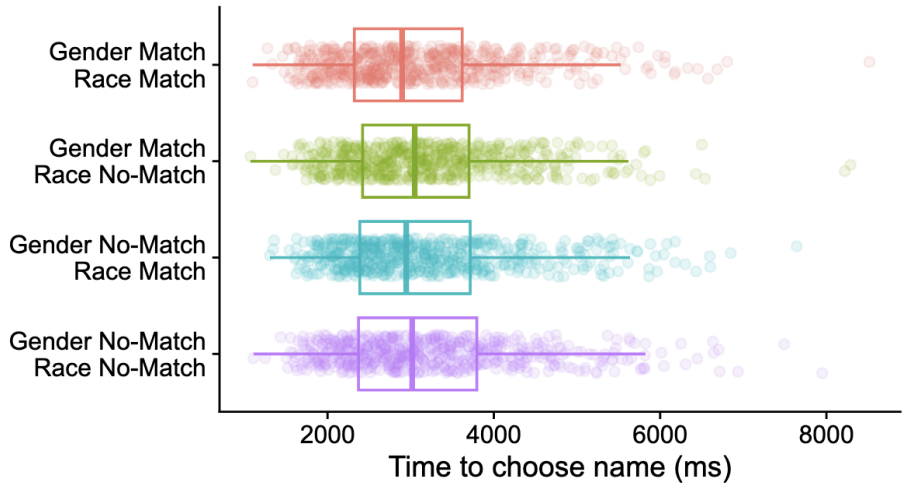


Fig. 2: Our own visualisation showing response time across the four conditions

Bearing in mind the outliers visible in Fig. 2, we reran the analysis with a more conservative cut-off point of 6 s than that of 10 s preregistered and applied by the authors (corresponding to 91% of the trials analysed by authors) to test the robustness of the reported effect. The differences in both observed and predicted mean reaction times across the three conditions are smaller than those reported, but remain nonetheless significant. We also reran the model on this subset of the data. The results remain broadly the same (see Table 7).

Table 7 – Results of robustness check on naming memory task analysis

Claim in the original study (as reported on p. 1709-1710)	Computational reproduction	Robustness check
Response times significantly increased as demographic match decreased	β = 0.020656	β = 0.017077
	(0.004641)	(0.003973)
	t(642) = 4.451	t(613) = 4.298
	p = 0.0000101	0.00002
Participants were fastest to select names when they matched the figures on both gender and race	2,724 ms	2,548 ms
	[2,642–2,801]	[2,488–2,612]
When participants matched the figures in gender but differed in race (and, thus, the forms of names were perfectly counterbalanced), then responses were about 90 ms slower	2813-2724 = 89 ms	2,639-2,548 = 91 ms
	[2,734–2,898]	[2,575–2,703]
When participants matched in race but differed in gender, then responses were about 70 ms slower	2,794-2,724 = 70 ms	2,605-2,548 = 57 ms
	[2,713–2,886]	[2,538–2,673]
Finally, when participants differed from the figures in both gender and race, they were an additional 50 ms slower to name them	2,856-2,794 = 62 ms	2,662-2,605 = 57 ms
	[2,769–2,946]	[2,595–2,728]
Model performance metrics		
N	22482	12311
Marginal R2 / Conditional R2	0.001 / 0.287	0.001 / 0.282

4. Conclusion

To conclude, although the authors generously provided both data and R code, a full computational reproduction of all published results was not possible. The absence of a README file and codebook, missing details in the paper's methods section, and difficulties in matching some of the preregistrations with the reported analyses complicated the reproduction process.

Before we began with the computational reproductions, we identified some notable deviations from the preregistrations. In Study 2, the preregistered models do not appear to have been run. In Study 3, the preregistered "exploratory models for each of the valence conditions" are not mentioned in either the code or the paper. Several preregistrations include mixed-effects regression models with "syntax" as a predictor among others (such as gender, race), but the models run as part of the analyses only include "syntax" (without gender or race). For the Empathy Rating Study, it is not clear whether the preregistered exclusion criteria have been applied. Finally, the preregistration of the Omnibus Study is missing; yet, these analyses – which substantiate the main claims of the paper – are not labelled as exploratory (unlike Study 4).

We were able to successfully reproduce the results of Studies 1–3 with the provided data and scripts, barring small differences in the 95% bootstrapped confidence intervals. For Study 4, we were unable to reproduce the processed data files from the raw data and the median time taken to input the critical sentence, due to a lack of details. We were, however, able to reproduce the results of Study 4 by writing the analysis script ourselves. The only difference we observed is likely the result of a typographical error: the authors report a positive *z*-value, whereas we obtained the same value but with a negative sign (z = -1.299). For the Empathy Rating Study, we were unable to reproduce the processed dataset from the preprocessing script, but we successfully recreated the dataframe by adapting the authors' preprocessing script. For the Agency Rating Study, we were unable to reproduce the first chi-square value. Some of the scripts, notably the analysis script for the memory naming task, are not

immediately reproducible as they repeat previous data processing steps and require manual adjustments. In spite of these issues, we successfully reproduced most of the reported statistics, as well as all of the figures.

Our robustness tests showed that running the preregistered models led to similar findings as reported in the paper, particularly for Study 3. For Study 2, our robustness check confirmed that Black women are less likely to show a 'like me' effect, but we also noted that Black men displayed the strongest 'like me' effect of all groups, which is not mentioned in the article. Our robustness check for Study 4 confirmed the authors' claim that the inclusion of neutral names led to a smaller 'like me' effect. For the memory naming task, the results of the paper were also found to be robust to our manipulations. Overall, the findings and claims of the papers are, with some limitations, reproducible and robust to our alternative specifications and the preregistered analyses.

This reproduction attempt taught us that providing data and preprocessing and analysis scripts does not suffice; it is also important to include a README file to facilitate navigation through the files and folders. To avoid confusion, multi-study papers need to ensure that each study is consistently labelled. Detailed script annotations are also necessary to be able to retrace the authors' preprocessing and analytical steps, especially for studies published in journals with strict word count limits that inevitably lead to incomplete methods sections. For optimal transparency, we suggest that authors make clear which findings are the outcome of preregistered hypotheses and which are the outcome of exploratory analyses.

References

Brough, J., Harris, L.T., Wu, S.H. *et al.* 2024. Cognitive causes of 'like me' race and gender biases in human language production. *Nat Hum Behav* 8, 1706–1715. https://doi.org/10.1038/s41562-024-01943-3

Appendix

The data and code that we used as part of our reproduction are available at the following GitHub repository: https://github.com/mariekluenter/like_me_study_reproduction.

This repository contains a copy of the original paper (`paper_like_me.pdf`), as well as the authors' original scripts and datasets for ease of access (`R_scripts/scripts_and_data_original/`). The folder (`R_scripts/scripts_reproduction/`) contains our modified scripts, based on the authors' originals. The folder structure and file names follow the same nomenclature as the original files with the suffix `_reproduction`.